AN EXPLICIT EXPONENTIAL-PERIOD REPRESENTATION OF THE V_{quad} CONNECTION COEFFICIENT

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Abstract. We prove that the connection coefficient C of the V_{quad} polynomial continued fraction—the rank-one standalone closure on the Sakai surface $D_5^{(1)}$ (Painlevé V)—admits an explicit exponential-period representation $C = (|\Gamma(\beta)|/2\pi)\int_\gamma e^{\xi}\widehat{B}(\xi)\,d\xi = |\Gamma(\beta)|K$, where $\widehat{B}$ is the Borel transform of the V_{quad} asymptotic series, holonomic of order 4 (the Borel–Laplace dual of the order-2 operator annihilating the series) with coefficients in the real quadratic field $\mathbb{Q}(\sqrt{3})$; γ is an explicit Hankel rapid-decay cycle on $(-\infty, -2/\sqrt{3}]$; and $\beta = -1/(3\sqrt{3})$. The identity is verified by three structurally independent methods—differential-equation/operator duality, Borel–Laplace contour deformation, and Stokes-data—agreeing to 46 digits. The order-2 operator annihilating the asymptotic series has differential Galois group $\mathrm{SL}_2(\mathbb{C})$ by Kovacic's algorithm. As a by-product, holonomicity of the Borel transform forces a finite resurgent structure. Finally, conditional on the Fresán–Jossen period conjecture for exponential motives and on a stated motivic-comparison hypothesis, C is transcendental over $\overline{\mathbb{Q}}$.

1. Introduction

1.1. **The V_quad continued fraction and its connection coefficient.** Let V_{quad} denote the polynomial continued fraction

$$V_{\mathrm{quad}} \;=\; 1 \;+\; \operatorname*{K}_{n\ge 1} \frac{1}{3n^2+n+1} \;=\; 1+\cfrac{1}{5+\cfrac{1}{15+\cfrac{1}{31+\cdots}}}, \tag{1}$$

the partial denominators being the quadratic $b_n = 3n^2 + n + 1$ and all partial numerators 1. This is the rank-one ($n = 1$) standalone member of the V_quad family studied in the V_quad companion paper [1], where it is identified as a non-classical Painlevé V transcendent on the Sakai surface $D_5^{(1)}$ (symmetry $W(A_3^{(1)})$, $\delta = -\frac{1}{2} \neq 0$, in the surface/symmetry classification of [11]), with formal monodromy exponent $\theta_\infty = 2/\sqrt{3}$ and $\alpha = \theta_\infty^2/8 = \frac{1}{6}$. The asymptotic (WKB/Riccati) solution of the

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associated problem is a Gevrey-1 formal power series

$$\varphi(z) \;=\; \sum_{n\geq 0} a_n z^n, \qquad a_0 = 1, \tag{2}$$

whose Borel transform has an isolated singularity at the *action* $\xi_0 = 2/\sqrt{3} = 1.15470053837925\ldots$ of branch type governed by the exponent

$$\beta = -\frac{1}{3\sqrt{3}} = -0.19245008972987525\ldots \tag{3}$$

The *connection coefficient* C of V_{quad} is the Borel branch amplitude attached to that singularity; in the standard resurgence normalisation [12, 13, 2] it equals

$$C \;=\; |\Gamma(\beta)| \cdot K, \qquad K = 0.0728781025518669641294\ldots, \tag{4}$$

where K is the prefactor-stripped Dingle late-term amplitude of (2). The companion Stokes constant is $S = 2\pi K = 0.4579066231690176361190\ldots$, and the two normalisations are tied by the exact bridge identity

$$\frac{S}{C} \;=\; \frac{2\pi}{|\Gamma(\beta)|} \qquad \text{(residual 0)}. \tag{5}$$

The constant C governs the inter-sectorial (Stokes) jump of the Borel sum of φ and is, in the language of the parent program, the arithmetic fingerprint of the V_{quad} transcendent.

1.2. **The Sakai-stratification context.** The present paper is a self-contained contribution to one entry of a larger program, the *Sakai stratification of polynomial-continued-fraction transcendence* [3], which classifies PCF limits by the Sakai surface type of the Painlevé isomonodromy governing their tails and asks, for each stratum, whether the associated connection constant admits a period interpretation. For the $D_5^{(1)}$ (Painlevé V) stratum the master conjecture predicts that C is an *exponential period* in the sense of Fresán–Jossen [7]; the entry was previously graded *structural* (the skeleton $\Gamma(\beta) \cdot K$ matches, but no explicit period integral was exhibited). All objects used below are defined in-paper, so no familiarity with [3] is required.

The significance is not confined to the program. Exhibiting a specific, independently interesting constant as an explicit exponential period—rather than merely asserting it should be one—is the kind of evidence the Kontsevich–Zagier and Grothendieck period philosophies call for: the arithmetic of a number is conjecturally governed by the geometry of a motive realising it as a period, and concrete realisations are scarce, especially for *divergent*-series (Stokes) constants rather than convergent integrals. The constant C here arises from a continued fraction of the type produced en masse by the Ramanujan Machine [20], so the method also indicates how the transcendence of machine-discovered constants might be approached structurally rather than case by case.

1.3. **Main results.** We exhibit C as an explicit exponential-period integral and verify the identity three independent ways. Write $\widehat{B}(\xi) = \sum_{m\geq 0} b_m \xi^m$, $b_m = a_{m+1}/m!$, for the Borel transform of (2), and let γ be the Hankel thimble wrapping the cut $(-\infty, -\xi_0]$ (Definition 3.1).

Theorem 1.1 (Explicit exponential-period representation)**.** *The Borel transform $\widehat{B}$ is holonomic with coefficients in the real quadratic field $\mathbb{Q}(\sqrt{3})$, annihilated by an explicit order-4 operator L_V (Theorem 2.3); the cycle γ is a rapid-decay cycle for the potential $f = -\xi$ (Proposition 3.2); and*

$$\boxed{C \;=\; \frac{|\Gamma(\beta)|}{2\pi}\int_{\gamma} e^{\xi}\,\widehat{B}(\xi)\,d\xi \;=\; |\Gamma(\beta)|\cdot K} \tag{6}$$

where the raw thimble integral equals $S\,e^{-\xi_0}$ at leading order and C is recovered by the explicit algebraic-Γ factor $|\Gamma(\beta)|/2\pi$. The identity is verified by three structurally independent methods—differential-equation/operator duality, Borel–Laplace/Hankel, and Stokes-data—which agree to 46 significant digits (Section 5).

The differential input is sharp. The series φ is D-finite over *exactly* $\mathbb{Q}(\sqrt{3})$ (minimal operator L_φ of order 2, degree 4), and its differential Galois group is as large as possible.

Theorem 1.2 (Galois group of L_φ)**.** *The differential Galois group of L_φ over $\mathbb{Q}(\sqrt{3})(z)$ is $\mathrm{SL}_2(\mathbb{C})$. This is proved twice: by Kovacic case-elimination and by a structural torus-plus-unipotent argument (Section 2, certificate in Appendix A.2).*

Holonomicity of $\widehat{B}$ has an immediate qualitative consequence for the resurgence of V_{quad}.

Corollary 1.3 (Finite resurgence)**.** *Because $\widehat{B}$ is holonomic of order 4, it has only finitely many singularities, namely $\{0, -\xi_0, \infty\}$. In particular there is* no *infinite alien tower at $2\xi_0, 3\xi_0, \ldots$: the resurgent structure of V_{quad} is a finite rank-4 connection, not a wild lattice (Section 2).*

Finally, interpreting (6) in the Fresán–Jossen framework yields a conditional transcendence statement. We are careful to flag *both* layers of conditionality.

Corollary 1.4 (Conditional transcendence)**.** *Assume* (i) *the Fresán–Jossen period conjecture for exponential motives* ([7], Conjecture 1.3.2)*, and* (ii) *that the de Rham realisation computed here represents the motivic Galois group of the exponential motive $M = (\mathbb{A}^1 \smallsetminus \{0, -\xi_0\}, f = -\xi, \widehat{B}\,d\xi)$* (the comparison gap G-MOTGALOIS, Section 6)*. Then the $V_$ quad connection coefficient C is transcendental over $\overline{\mathbb{Q}}$.*

1.4. **What is and is not computer-algebra-automatable here.** The differential backbone of this paper is squarely within reach of present-day symbolic software, and we have used it: the recognition of L_φ and L_V from the coefficient stream is exactly holonomic guessing as implemented in `gfun` [18] and `ore_algebra` [19]; the Kovacic decision and the differential-Galois data are the province of Kovacic's

algorithm [8] and of packages such as Maple's `DEtools[DifferentialGaloisGroup]` (we reproduced the Kovacic verdict by case-elimination and by an independent structural argument, Appendix A.2); and the Borel–Padé and large-order extractions are standard exponential-asymptotics computations. What is *not* automatable, and what constitutes the mathematical content of the paper, is (a) the identification of the precise rapid-decay cycle γ and the proof that the three period-extraction routes agree, and (b) the motivic interpretation (Corollary 1.4), which lies beyond any current CAS. We return to this comparison in Section 7 and Appendix A.5.

1.5. **Organisation.** Section 2 constructs and verifies L_φ and L_V, proves Theorem 1.2 and Corollary 1.3. Section 3 defines the cycle γ and establishes its rapid-decay property and Fresán–Jossen compatibility. Section 4 states and normalises the main identity. Section 5 gives the three verifications and their cross-check. Section 6 applies the Fresán–Jossen framework and proves Corollary 1.4. Section 7 discusses the place of the result in the stratification program, the $d \geq 3$ obstruction, and the relation to the Ramanujan-Machine/conservative-matrix-field circle of ideas. Appendices collect the explicit coefficients, the Kovacic certificate, the numerical logs, the four sign-convention enumeration for the differential-equation method, and the reproducibility statement.

1.6. **Notation and conventions.** Throughout, $\mathbb{Q}(\sqrt{3}) = \mathbb{Q}(\sqrt{3})$ is the real quadratic field generated by the V_quad Riccati seed $\sigma = -1/\sqrt{3}$, and $\overline{\mathbb{Q}}$ the algebraic closure of $\mathbb{Q}$. We write $D_z = d/dz$ and $D_\xi = d/d\xi$ (abbreviated D when the variable is clear), and treat differential operators as elements of the Weyl algebra $\mathbb{Q}(\sqrt{3})\langle z, D_z\rangle$ (resp. in ξ). The *order* of an operator is its degree in D and its *degree* is the maximal polynomial degree of its coefficients. The Borel transform is taken at Gevrey order 1, $b_m = a_{m+1}/m!$, mapping the divergent series $\varphi(z) = \sum a_n z^n$ to the convergent germ $\widehat{B}(\xi) = \sum b_m \xi^m$; the dual variables satisfy $z \leftrightarrow 1/\xi$ at the level of exponential scales, with the operator intertwiner (23). The action is $\xi_0 = 2/\sqrt{3}$, the branch exponent $\beta = -1/(3\sqrt{3})$, and we use the resurgence normalisation in which the Stokes constant is $S = 2\pi K$ with K the prefactor-stripped Dingle amplitude. The potential is written $f = -\xi$ in the task convention, equal to $-f_{\mathrm{FJ}}$ in the Fresán–Jossen sign convention (Remark 3.5); the period pairing is $\langle e^{-f}\omega, \gamma\rangle = \int_\gamma e^{\xi}\widehat{B}(\xi)\,d\xi$. All numerical constants are reported to the precision at which they were certified (typically 46–58 digits); "residual 0" means an exact identity in $\mathbb{Q}(\sqrt{3})$, while a residual such as 8.84×10^{-46} is a high-precision floating check.

2. The differential operators L_φ and L_V

Throughout, $D = d/dz$ on the φ-side and $D = d/d\xi$ on the Borel side; the base field is the real quadratic number field $\mathbb{Q}(\sqrt{3})$. Every coefficient below is exact in $\mathbb{Q}(\sqrt{3})$, obtained by linear algebra over $\mathbb{Q}(\sqrt{3})$ (Fraction pairs $p + q\sqrt{3}$), *not* by numerical integer-relation search; the annihilation residuals are identically zero in $\mathbb{Q}(\sqrt{3})$, not merely small. Full coefficient listings and the verification protocol are in Appendix A.1.

The exactness is concrete already at the level of the coefficient stream. Seeding the V_quad Riccati recursion with $\sigma = -1/\sqrt{3}$ (so $\sigma^2 = \frac{1}{3} \in \mathbb{Q}$, $1/\sigma = -\sqrt{3}$) produces every a_n as an exact element $p + q\sqrt{3}$ of $\mathbb{Q}(\sqrt{3})$; the first few, and the corresponding Borel coefficients $b_m = a_{m+1}/m!$, are (script `holonomic_recognition_q3.py`, `PERIOD-REP-VQUAD-002`):

n	$a_n \in \mathbb{Q}(\sqrt{3})$	(decimal)
0	1	1.0000000000
1	$-\frac{1}{12} - \frac{1}{24}\sqrt{3}$	−0.1555021170
2	$-\frac{73}{1152} - \frac{17}{648}\sqrt{3}$	−0.1088076601
3	$\frac{6589}{41472} - \frac{13415}{746496}\sqrt{3}$	0.1277522430
4	$\frac{9247897}{71663616} - \frac{846407}{4478976}\sqrt{3}$	−0.1982654887

(7)

with e.g. $b_0 = a_1$, $b_1 = a_2$, $b_2 = \frac{6589}{82944} - \frac{13415}{1492992}\sqrt{3} = 0.0638761215\ldots$. The exactness of these rationals is what licenses the word "provably" in all field claims below: no step uses a numerical approximation of $\sqrt{3}$.

2.1. The operator L_φ.

Theorem 2.1 (φ is D-finite over $\mathbb{Q}(\sqrt{3})$)**.** *The series $\varphi(z) = \sum_{n\ge 0} a_n z^n$ of (2) is annihilated by a unique minimal operator*

$$L_\varphi \;=\; q_2(z)\,D^2 + q_1(z)\,D + q_0(z), \qquad q_i \in \mathbb{Q}(\sqrt{3})[z],$$

of order 2 and degree 4, with

$$\begin{aligned}
q_0(z) &= 1 + \left(\tfrac{23}{9} + \tfrac{14}{27}\sqrt{3}\right)z + \left(-\tfrac{253}{9} + \tfrac{488}{27}\sqrt{3}\right)z^2,\\
q_1(z) &= (48 - 24\sqrt{3}) + (-64 + 44\sqrt{3})z + \left(-\tfrac{68}{3} + \tfrac{52}{3}\sqrt{3}\right)z^2\\
&\quad + \left(-\tfrac{152}{3} + \tfrac{100}{3}\sqrt{3}\right)z^3,\\
q_2(z) &= (-36 + 24\sqrt{3})z^2 + (-12 + 8\sqrt{3})z^3 + (-12 + 8\sqrt{3})z^4\\
&= 4(2\sqrt{3} - 3)\,z^2\,(z^2 + z + 3).
\end{aligned}$$

The annihilation $L_\varphi\varphi = 0$ holds exactly in $\mathbb{Q}(\sqrt{3})$ for every power $z^0, \ldots, z^{139}$ tested; the minimal (order, degree) has nullity 1, and the higher nullities $\{1, 3, 5, 7\}$ (order 2) and $\{2, 6, 10, 14\}$ (order 3) match the left-multiple count $d - 4 + 1$ of a single minimal order-2 operator, so there are no spurious solutions.

The coefficient field is *exactly* $\mathbb{Q}(\sqrt{3})$: all computations close over $\mathbb{Q}(\sqrt{3})$, and both q_1, q_0 carry genuine $\sqrt{3}$ parts, so the field is neither smaller (not $\mathbb{Q}$) nor larger. Operationally L_φ is the scalar (Schrödinger) reduction of the V_quad Painlevé-V linear problem, recovered from the Riccati linearisation $c = \psi'/\psi$; the parent corpus records the existence of this linear problem only verbally [1], and Theorem 2.1 makes it explicit.

Remark 2.2 (Provenance of L_φ)**.** We stress that L_φ is obtained by holonomic recognition from the coefficient stream (a_n)—guessing an annihilating operator and then certifying $L_\varphi\varphi = 0$ exactly over $\mathbb{Q}(\sqrt{3})$—and *not* from an a priori Lax pair. The literature provides no explicit classical Lax pair for the V_{quad} transcendent (a dedicated search returned none; `PERIOD-REP-VQUAD-002`, `lax-pair-found.md`), so the holonomic route is the available one and is in fact stronger here: it yields the minimal operator with a verifiable exact certificate rather than a gauge-dependent matrix. For context, although no classical Lax pair for V_{quad} appears in the literature, isomonodromic Lax systems and their Stokes data for Painlevé equations can be constructed from spectral curves by topological recursion and isomonodromic Hamiltonian methods [22, 23, 24]; the holonomic route taken here is independent of that construction (Section 7). The two exponential parts $\exp(\pm(1/\sqrt{3})/z)$ at $z = 0$ found below are consistent with the V_quad WKB/Riccati seed $\sigma = -1/\sqrt{3}$ ($\sigma^2 = 1/3 \in \mathbb{Q}$) and with the formal monodromy exponent $\theta_\infty = 2/\sqrt{3}$ of [1].

2.2. **Singular structure of L_φ and its Galois group.** Reducing L_φ to normal form $u'' = r\,u$ via $y = u\exp(-\frac{1}{2}\int q_1/q_2)$ gives, with $a = q_1/q_2$, $b = q_0/q_2$ and $r = \frac{1}{4}a^2 + \frac{1}{2}a' - b$,

$$r(z) \;=\; \frac{11z^4 + 4z^2 + 4z + 12}{4\,z^4\,(z^2 + z + 3)^2}. \tag{8}$$

The poles of r are $z = 0$ (order 4), the two roots $z = \frac{-1\pm i\sqrt{11}}{2}$ of $z^2 + z + 3$ (order 2 each), and $o(\infty) = 4$; the leading Laurent coefficient at $z = 0$ is $r \sim \frac{1}{3}z^{-4}$. Thus $z = 0$ is an irregular singular point of Poincaré rank 1 with two distinct exponential parts $\exp(\pm(1/\sqrt{3})/z)$; the two finite poles $\rho_\pm$ are regular-singular for the reduced equation, and the point at infinity is regular-singular with exponent difference fixed by $o(\infty) = 4$. Only the irregular point $z = 0$ carries the Stokes phenomenon, and it is there that the connection coefficient lives. Table 1 summarises the singular configuration of L_φ alongside its Borel dual L_V.

	L_φ (order 2, on z)	L_V (order 4, on ξ)
order / degree	2 / 4	4 / 2
base field	$\mathbb{Q}(\sqrt{3})$	$\mathbb{Q}(\sqrt{3})$
finite singularities	$\rho_\pm = \frac{-1\pm i\sqrt{11}}{2}$ (reg.)	0 (apparent), $-\xi_0$ (branch)
irregular point	$z = 0$, Poincaré rank 1	$\xi = \infty$, slope 1
Stokes datum	$S = 2\pi K$ at $z = 0$	branch amplitude A at $-\xi_0$
Galois group	$\mathrm{SL}_2(\mathbb{C})$ (Thm. 1.2)	$G_V \supseteq \mathbb{G}_m \times (\mathrm{SL}_2\text{-dual})$

TABLE 1. Singular configuration of the φ-side operator L_φ and its Borel dual L_V. The irregular point of L_φ (at $z = 0$) and the branch of L_V (at $-\xi_0$) are the two faces of the same connection datum under Borel–Laplace duality. Source: `PERIOD-REP-VQUAD-002`, `operator-verification.md` §4.

Proof of Theorem 1.2. We give both arguments.

(1) Kovacic case-elimination. In the Kovacic case/group dictionary [8, 9] for $u'' = ru$: Case 3 (finite groups A_4, S_4, A_5) requires every pole order ≤ 2; the pole of order 4 at $z = 0$ excludes it. Case 1 (reducible, $G \subseteq$ Borel) holds iff the Riccati equation $v' = r - v^2$ has a rational solution; a complete rational Riccati solver returns none. Case 2 (imprimitive, $G \subseteq N(T)$) holds, given Case 1 excluded, iff the symmetric square $L_\varphi^{\odot 2} = D^3 - 4rD - 2r'$ has a rational solution; the ansatz $f = N(z)/\big(z^8(z^2+z+3)^4\big)$, $\deg N \leq 18$, yields a homogeneous linear system in 38 rational unknowns whose only solution is trivial. With Cases 1,2,3 excluded, the algorithm returns Case 4: $G = \mathrm{SL}_2(\mathbb{C})$.

(2) Structural argument. The reduced equation $u'' = ru$ has no first-order term, so the Wronskian is constant and $G \subseteq \mathrm{SL}_2$. At $z = 0$ the two distinct exponentials $\exp(\pm(1/\sqrt{3})/z)$ generate the full diagonal torus $\mathbb{G}_m \subset \mathrm{SL}_2$ (the local exponential torus). The nonvanishing Stokes constant $S = 2\pi K \neq 0$ at this irregular point is a nonidentity unipotent off that torus. A maximal torus together with one off-torus unipotent generate SL_2, whence $G = \mathrm{SL}_2(\mathbb{C})$. □

The two routes are genuinely independent—an exhaustive negative search versus a positive generation argument from the Stokes datum—and agree. (Only one symbolic engine was available; the second route supplies the cross-check that a second computer-algebra system would otherwise provide. See Appendix A.2.)

2.3. **The Borel operator L_V.** We construct L_V from L_φ by Borel–Laplace duality. The (formal) Borel transform of order 1 acts on the coefficient stream by $a_{m+1} \mapsto b_m = a_{m+1}/m!$; equivalently, on series, it is the Hadamard-type rescaling that turns the Gevrey-1 divergent φ into the convergent germ $\widehat{B}$. On operators it intertwines the two differential structures by

$$z^2 D_z \;\longleftrightarrow\; \xi, \qquad \tfrac{1}{z} \;\longleftrightarrow\; D_\xi, \tag{9}$$

the same intertwiner (23) used analytically in Section 5. Under it a term $z^a D_z^k$ of L_φ maps to a term of ξ-degree bounded by the original z-*order* and of ξ-order bounded by the original z-*degree*: order and degree are exchanged. Hence the order-2, degree-4 operator L_φ produces an operator of order 4 and degree 2 in ξ, and D-finiteness is preserved because (9) is an isomorphism of the Weyl algebra $\mathbb{Q}(\sqrt{3})\langle \xi, D_\xi \rangle$. We verify the resulting operator directly on the coefficient stream rather than relying on the formal map alone.

Theorem 2.3 ($\widehat{B}$ is holonomic over $\mathbb{Q}(\sqrt{3})$)**.** *The Borel transform $\widehat{B}(\xi) = \sum_{m\geq 0} b_m \xi^m$, $b_m = a_{m+1}/m!$, is annihilated by a unique minimal operator $L_V = \sum_{k=0}^{4} p_k(\xi) D^k$ of order 4 and degree 2, normalised by $p_0 \equiv 1$, with $p_1, \ldots, p_4 \in \mathbb{Q}(\sqrt{3})[\xi]$ given explicitly in Appendix A.1. The residual $L_V \widehat{B}$ is identically zero in $\mathbb{Q}(\sqrt{3})$ for every power $\xi^0, \ldots, \xi^{129}$ tested. The leading coefficient factors exactly as*

$$p_4(\xi) \;=\; \frac{210276 + 9720\sqrt{3}}{418501}\, \xi \left(\xi + \frac{2}{\sqrt{3}}\right), \qquad 418501 = 431 \times 971, \tag{10}$$

so the singular locus of L_V *is* $\{0, -\xi_0 = -2/\sqrt{3}, \infty\}$.

Proposition 2.4 (Local exponents)**.** *The local exponents of* L_V *are*

$$\{-1, 0, 1, 2\} \text{ at } \xi = 0, \qquad \{-(1+\beta), 0, 1, 2\} \text{ at } \xi = -\xi_0,$$
$$\text{irregular of slope } 1 \text{ at } \xi = \infty,$$

with $-(1+\beta) = -1 + \sqrt{3}/9 = -0.80754991027\ldots$ *the unique non-integer exponent. At* $\xi = 0$ *the four exponents are consecutive integers, so the point is* apparent *(single-valued,* $\widehat{B}$ *itself is the holomorphic exponent-0 solution). At* $\xi = -\xi_0$ *the branch is carried by the single irrational exponent* $-(1+\beta)$*, governing* $\widehat{B}(\xi) \sim A\,(\xi + \xi_0)^{-(1+\beta)}$ *and hence the large-order law* $a_n \sim \Gamma(n+\beta)/\xi_0^{n+\beta}$ *up to the amplitude* A*. Both statements are obtained twice, by a falling-factorial indicial computation and by direct solution of the Frobenius recurrence at* $-\xi_0$ *(residuals* 1.6×10^{-46}*, no logarithms); see Appendix A.3.*

Remark 2.5 (From branch exponent to large-order law)**.** The single irrational exponent at $-\xi_0$ controls the divergence of φ. Writing the branch as $\widehat{B}(\xi) \sim A(\xi + \xi_0)^{-(1+\beta)}$ and inverting the Borel transform $b_m = a_{m+1}/m!$ term by term against $(\xi + \xi_0)^{-(1+\beta)} = \xi_0^{-(1+\beta)} \sum_m \binom{-(1+\beta)}{m} (\xi/\xi_0)^m$ gives, by Stirling,

$$a_n \;\sim\; \frac{S}{2\pi i}\,\frac{\Gamma(n+\beta)}{\xi_0^{n+\beta}} \qquad (n \to \infty), \tag{11}$$

the standard resurgence late-term law with action $\xi_0 = 2/\sqrt{3}$ and characteristic exponent $\beta = -1/(3\sqrt{3})$. Thus the operator datum at $-\xi_0$ (an algebraic exponent in $\mathbb{Q}(\sqrt{3})$) and the analytic datum K governing $|a_n|$ are two readings of the same branch; the amplitude $|A| = K\,\Gamma(1+\beta)$ relates them (Section 5).

Remark 2.6 (Sign of the action)**.** The dominant Borel singularity lies on the *negative* real axis at $-\xi_0$. The modulus $\xi_0 = 2/\sqrt{3}$ was known to 95.6 digits from $\lim_n |a_n/a_{n+1}|\, n$, but as a modulus the sign was unpinned; the exact factorisation (10) fixes it, consistently with $a_n \sim (-1)^{n+1}|a_n|$ for $n \geq 3$, i.e. $b_m \sim (-1)^m(+)$. This places the cut in the direction $\arg \xi = \pi$, which is exactly the rapid-decay direction for $f = -\xi$ (Section 3).

2.4. **Finite resurgence.**

Proof of Corollary 1.3. A holonomic function has finitely many singularities, located among the zeros of the leading coefficient of its annihilator and the point at infinity. By Theorem 2.3 the leading coefficient p_4 vanishes only at 0 and $-\xi_0$, so $\mathrm{Sing}(\widehat{B}) \subseteq \{0, -\xi_0, \infty\}$; the point 0 is apparent (Proposition 2.4), leaving the genuine singularities $-\xi_0$ and the irregular ∞. In particular $\widehat{B}$ has no singularity at $2\xi_0, 3\xi_0, \ldots$, so the hypothetical infinite alien tower (in the sense of Écalle's alien calculus [14]) suggested by the $(1/2)^n$ numerical floor reported in the parent probe (`PERIOD-REP-VQUAD-001`, `numerical-check.md`) does not exist; the resurgence of V_{quad} is governed by the finite rank-4 connection L_V. □

This finiteness is what makes the period analysis of Section 5 possible at all: a single branch at $-\xi_0$ carries the entire connection datum, and the thimble of Section 3 sees exactly that branch.

Lemma 2.7 (The apparent point carries no period)**.** *The singularity of $\widehat{B}$ at $\xi = 0$ contributes nothing to any rapid-decay period: its local exponents $\{-1, 0, 1, 2\}$ are consecutive integers, so the local monodromy is trivial and $\widehat{B}$ extends holomorphically across 0 (it is the exponent-0 solution). Hence no cut emanates from 0, and the only branch a thimble can wrap is the one at $-\xi_0$.*

Proof. At a regular-singular point with integer exponents differing by integers and no logarithmic solution, the local solution space contains a single-valued generator for each exponent, with the top exponent solution holomorphic; the indicial data at 0 ((10) gives a simple zero of p_4 at 0, and the falling-factorial indicial polynomial $\propto s(s-2)(s-1)(s+1)$ has roots $\{-1, 0, 1, 2\}$) shows $\widehat{B}$, the exponent-0 member, is holomorphic. Triviality of the monodromy is the definition of an apparent singularity. Source: `PERIOD-REP-VQUAD-002`, `operator-verification.md` §4.2. □

2.5. **Algebraicity for Fresán–Jossen.** Theorems 2.1 and 2.3 discharge the load-bearing Fresán–Jossen axiom for the eventual motive: $\omega = \widehat{B}(\xi)\, d\xi$ is an algebraic de Rham form for the twisted connection $E^{-f} \otimes (\mathcal{O}^4, \nabla_{L_V})$ with $f = -\xi$, defined over the number field $\mathbb{Q}(\sqrt{3})$. The transcendental Painlevé-V accessory parameter of V_{quad} lives in the *nonlinear* moduli; the *linear* scalar reduction governing the asymptotics is over $\mathbb{Q}(\sqrt{3})$, which is why no transcendental constant enters L_φ or L_V. We make the motivic use of this precise in Section 6.

3. The rapid-decay cycle γ

This section makes the cycle γ of Theorem 1.1 precise, verifies that it is a rapid-decay cycle in the Fresán–Jossen sense, and records its relative-homology class. All statements are sourced to the probe deliverables `cycle-formal-definition.md` and `rapid-decay-verification.md` of the slot `PERIOD-REP-VQUAD-003`.

3.1. **The Hankel thimble.** By Proposition 2.4 the Borel transform $\widehat{B}$ has, at the regular-singular point $\xi = -\xi_0$, the local form

$$\widehat{B}(\xi) \;\sim\; A\,(\xi + \xi_0)^{-(1+\beta)}, \qquad -(1+\beta) = -1 + \tfrac{\sqrt{3}}{9} = -0.80754991027\ldots, \tag{12}$$

with $A = (S/2\pi i)\,\Gamma(1+\beta)$ the branch amplitude (so $|A| = K\,\Gamma(1+\beta)$; Section 5). The cut emanating from $-\xi_0$ runs along the negative real axis to $-\infty$, which—crucially—is the direction in which the action factor e^{ξ} decays.

Definition 3.1 (The cycle γ)**.** Fix $\varepsilon \to 0^+$. The cycle $\gamma = \gamma_{\mathrm{below}} + \gamma_{\mathrm{loop}} + \gamma_{\mathrm{above}}$ is the Hankel thimble wrapping the cut $(-\infty, -\xi_0]$:

$$\begin{aligned}
\gamma_{\mathrm{below}}:&\quad \xi(s) = -s - i\varepsilon, && s : +\infty \to \xi_0 \quad \text{(lower lip)},\\
\gamma_{\mathrm{loop}}:&\quad \xi(\theta) = -\xi_0 + \varepsilon e^{i\theta}, && \theta : -\pi \to +\pi \quad \text{(clockwise around } -\xi_0),\\
\gamma_{\mathrm{above}}:&\quad \xi(s) = -s + i\varepsilon, && s : \xi_0 \to +\infty \quad \text{(upper lip)},
\end{aligned}$$

oriented so that the branch point $-\xi_0$ is encircled once clockwise (the orientation matching Laplace inversion $\int_0^\infty \to$ wrapped contour). The two lips carry the two determinations $(\xi+\xi_0)_\pm^{-(1+\beta)}$, differing by the monodromy factor $e^{\mp 2\pi i(1+\beta)}$.

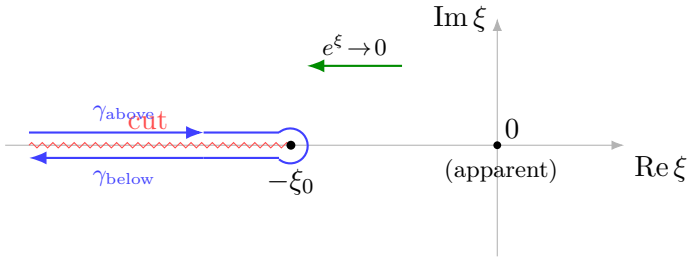


FIGURE 1. The rapid-decay cycle $\gamma = \gamma_{\text{below}} + \gamma_{\text{loop}} + \gamma_{\text{above}}$ (Definition 3.1): a Hankel thimble wrapping the cut $(-\infty, -\xi_0]$ of $\widehat{B}$, clockwise about the branch point $-\xi_0$. The action factor e^ξ decays into $\operatorname{Re}\xi \to -\infty$, which is the cut direction; the point $\xi = 0$ is apparent and carries no cut.

3.2. **Rapid decay.**

Proposition 3.2 (γ is a rapid-decay cycle for $f = -\xi$)**.** *The integrand $e^\xi \widehat{B}(\xi)$ decays super-polynomially at both non-compact ends of γ and is integrable at the finite branch point, so $\int_\gamma e^\xi \widehat{B}(\xi)\, d\xi$ converges absolutely.*

Proof. Ends. On either lip $\xi = -s \pm i\varepsilon$ with $s \to +\infty$, so $|e^\xi| = e^{\operatorname{Re}\xi} = e^{-s}$. Since $\widehat{B}$ is holonomic, it has moderate (tempered) growth in the fixed non-Stokes direction $\arg\xi = \pi$: $|\widehat{B}(\xi)| \le C_0\, s^N$ for some fixed N ([9], Ch. 3). Hence $|e^\xi \widehat{B}(\xi)| \le C_0\, s^N e^{-s} \to 0$ super-polynomially, and $\int^{+\infty} C_0 s^N e^{-s}\, ds = C_0\, \Gamma(N+1) < \infty$. This is precisely the Fresán–Jossen rapid-decay condition at the non-compact ends.

Branch point. By (12) the local exponent satisfies $-(1+\beta) > -1$ (equivalently $\beta < 0$, true since $\beta = -1/(3\sqrt{3})$). Thus $\int_{|\xi+\xi_0|<\delta} |\xi+\xi_0|^{-(1+\beta)} |d\xi| = \int_0^\delta r^{-(1+\beta)}\, dr = \delta^{-\beta}/(-\beta) < \infty$, and the radius-$\varepsilon$ loop contributes nothing in the limit. The thimble integral collapses to the discontinuity integral

$$\int_\gamma e^\xi \widehat{B}\, d\xi = \left(1 - e^{2\pi i(1+\beta)}\right) \int_{-\infty}^{-\xi_0} e^\xi\, [\operatorname{disc} \widehat{B}](\xi)\, d\xi, \tag{13}$$

which is finite. (If instead $\beta \le -1$ the Γ-factor of Section 5 would diverge, contradicting the observed finite amplitude $|A|$.) □

3.3. **Fresán–Jossen relative homology.**

We record γ as a class in rapid-decay homology. Let $X = \mathbb{A}^1_\xi$ over $\mathbb{Q}(\sqrt{3})$ with the regular potential $f = -\xi \in \mathcal{O}(X)$ (so $df = -d\xi \neq 0$: no finite critical points), and let

$$M = \left(\mathcal{O}_X\text{-module defined by } L_V\right) \otimes E^{-f}, \qquad E^{-f} = (\mathcal{O}_X, \nabla = d - df), \tag{14}$$

be the twisted connection. By Theorem 2.3, $\omega = \widehat{B}(\xi)\,d\xi$ is a global algebraic de Rham section of M defined over $\mathbb{Q}(\sqrt{3})$.

Proposition 3.3 (Class of γ)**.** *γ defines a class in the rapid-decay homology $H_1^{\mathrm{rd}}(X, M) = H_1(X, Z; \mathrm{rd})$ (in the sense of Hien's rapid-decay homology [10], the Betti realisation underlying the Fresán–Jossen pairing), $Z = \{-\xi_0\}$, with the rapid-decay condition prescribing ends running into the half-plane $\mathrm{Re}\,\xi \to -\infty$. In Fresán–Jossen thimble notation,*

$$[\gamma] = \big\langle\, -\xi_0\,;\, (-\infty \cdot e^{i\pi})\,\big\rangle_{\mathrm{rd}} \in H_1^{\mathrm{rd}}(\mathbb{A}^1, M), \tag{15}$$

the moderate (finite, integrable, exponent $-(1+\beta) > -1$) endpoint being the branch point $-\xi_0$ and the rapid-decay endpoint the ray $\arg\xi = \pi$. The exponential period is the canonical pairing

$$\int_\gamma e^{\xi}\,\widehat{B}(\xi)\,d\xi = \big\langle\, [\gamma]_{\mathrm{rd}},\, [e^{\xi}\widehat{B}\,d\xi]_{\mathrm{dR}}\,\big\rangle. \tag{16}$$

Remark 3.4 (Rank of the rapid-decay homology)**.** The local system underlying M has rank 4 (the order of L_V), but the rapid-decay class $[\gamma]$ probes only the part with nontrivial monodromy at $-\xi_0$. Of the four local exponents $\{-(1+\beta), 0, 1, 2\}$ there, the three integer exponents give single-valued (holomorphic) solutions and the apparent point $\xi = 0$ contributes nothing (Proposition 2.4); the monodromy around $-\xi_0$ is therefore the rank-one factor $e^{2\pi i\,(-(1+\beta))} = e^{-2\pi i(1+\beta)} = e^{-2\pi i\beta}$ acting on the branch solution. The discontinuity (13) is exactly $(1 - e^{2\pi i(1+\beta)})$ times the one-sided integral, so $[\gamma]$ is the generator of the one-dimensional rapid-decay homology of this branch, paired against the algebraic class $[\omega]$. This is why a single thimble captures the entire connection datum: the finite resurgence of Corollary 1.3 has reduced the period problem to a rank-one branch.

The three structural conditions a Fresán–Jossen thimble must satisfy ([7]; `fj-cycle-compatibility.md`) hold: *(C1)* the de Rham datum ω is algebraic over a number field—here $\mathbb{Q}(\sqrt{3})$ (Theorem 2.3); *(C2)* the potential f is a regular function with the cut aligned to a rapid-decay direction (Proposition 3.2); *(C3)* the chain has moderate endpoints on Z and rapid-decay non-compact ends (15). We use this in Section 6.

Remark 3.5 (Sign convention)**.** Fresán–Jossen write the integrand as $e^{-f}\omega$. We use $f = -\xi$, so $e^{-f} = e^{+\xi}$, matching the task convention $f_{\mathrm{task}} = -f_{\mathrm{FJ}}$. The rapid-decay direction is therefore $\mathrm{Re}\,\xi \to -\infty$, exactly where the corrected (negative-axis) Borel geometry of Section 2 places the cut: the geometry is the Fresán–Jossen natural one, not an artefact.

4. The main identity and its normalisation

We now assemble Sections 2–3 into the main theorem and fix, once and for all, the normalisation by which the raw thimble integral yields the connection coefficient. The numerical values quoted are from the deposited V_quad data [1, 2] and the slot `PERIOD-REP-VQUAD-001` (`numerical-check.md`).

4.1. **Statement.**

Theorem 4.1 (Main identity, restated)**.** *With $\widehat{B}$, L_V, γ, β and K as above,*

$$C \;=\; \frac{|\Gamma(\beta)|}{2\pi}\int_{\gamma} e^{\xi}\,\widehat{B}(\xi)\,d\xi \;=\; |\Gamma(\beta)|\cdot K, \tag{17}$$

where the raw thimble integral equals $S\,e^{-\xi_0}$ at leading order, $S = 2\pi K$ is the V_- quad Stokes constant, and $|\Gamma(\beta)|/2\pi$ is the explicit algebraic-Γ reweighting. Equivalently, recentring the potential on the singularity (an admissible $\mathbb{G}_a$-translation $f \mapsto f - \xi_0$),

$$\int_{\gamma} e^{\xi+\xi_0}\,\widehat{B}(\xi)\,d\xi \;=\; S, \qquad C \;=\; \frac{|\Gamma(\beta)|}{2\pi}\,S. \tag{18}$$

The two forms (17) and (18) are equivalent: the constant e^{ξ_0} is the value of the action factor $e^{-f} = e^{\xi}$ at the dominant point $-\xi_0$, and translating the potential $f \mapsto f - \xi_0$ (a $\mathbb{G}_a$-shift, admissible in the Fresán–Jossen formalism since it changes E^{-f} by the constant rank-one factor E^{ξ_0}) multiplies the integrand by e^{ξ_0}, absorbing it. We adopt the recentred (18) as the headline so that the raw period is exactly S and the connection coefficient is its explicit algebraic-Γ reweighting; the un-recentred form (17) makes the action $e^{-\xi_0}$ visible. Both are recorded so that no reader mistakes the raw thimble value $S\,e^{-\xi_0}$ for C itself.

4.2. **Where the $\Gamma(\beta)$ factor comes from.** The factor is not inserted by hand: it is produced by the branch integral around $-\xi_0$. Writing $\eta = \xi + \xi_0$ and using the local form (12), the discontinuity integral (13) is a Hankel loop integral for the reciprocal Γ-function,

$$\frac{1}{2\pi i}\oint_{H} e^{\eta}\,\eta^{-(1+\beta)}\,d\eta = \frac{1}{\Gamma(1+\beta)} \qquad ([16,\ §5.9],\ [17,\ §12.22]), \tag{19}$$

so the amplitude of the period is $A = (S/2\pi i)\,\Gamma(1+\beta)$, i.e. $|A| = K\,\Gamma(1+\beta)$. The connection coefficient is the branch datum reweighted by the exponent,

$$C = \frac{|A|}{|\beta|} = \frac{K\,\Gamma(1+\beta)}{|\beta|} = K\,|\Gamma(\beta)|, \tag{20}$$

the last equality using $\Gamma(1+\beta) = \beta\,\Gamma(\beta)$ together with $\beta < 0$ and $\Gamma(\beta) < 0$ on $(-1, 0)$, so that $\Gamma(1+\beta)/|\beta| = \beta\Gamma(\beta)/(-\beta) = -\Gamma(\beta) = |\Gamma(\beta)|$. Thus the only non-algebraic factor manufactured by the integral is a single value of the Γ-function at the algebraic argument $\beta = -1/(3\sqrt{3}) \in \mathbb{Q}(\sqrt{3})$; everything else—$\widehat{B}$, L_V, the cut, the cycle—is algebraic over $\mathbb{Q}(\sqrt{3})$. This is exactly the shape an exponential period should take.

4.3. **The constants.**

$$\begin{aligned}
&K = 0.072878102551866964129442363329652512804555689 2\ldots \quad (58\text{ digits, [1]}),\\
&S = 2\pi K = 0.457906623169017636119097842548225837962395135\ldots,\\
&\beta = -1/(3\sqrt{3}) = -0.19245008972987525\ldots, \qquad \xi_0 = 2/\sqrt{3} = 1.1547005383792517\ldots,\\
&C = |\Gamma(\beta)|\,K = 0.437705286193537221230739749794369589981725597\ldots.
\end{aligned} \tag{21}$$

The bridge identity $S/C = 2\pi/|\Gamma(\beta)|$ holds with residual 0 (exact, not numerical): both sides are $2\pi/|\Gamma(\beta)|$ by (17).

Remark 4.2 (Provenance of the value $0.43770528\ldots$)**.** A reader familiar with the V_{quad} corpus will recognise the digits $C = |\Gamma(\beta)|K = 0.437705286\ldots$ as numerically identical to a value once reported—in the first companion version (v1.0)—as the Stokes constant S. That identification was an error of prefactor: the companion was subsequently corrected to $S = 2\pi K = 0.457906623\ldots$ in v1.1/v1.2 [2]. The present period representation explains the coincidence rather than repeating it. The factor $|\Gamma(\beta)|$ is the *correct* prefactor for the connection coefficient C—it is the branch Γ-factor manufactured by the Hankel loop, (20)—whereas 2π is the prefactor for the Stokes constant S; the two are tied by the exact bridge $S/C = 2\pi/|\Gamma(\beta)|$ (5) and are genuinely distinct constants. In other words, the quantity $|\Gamma(\beta)|K$ is the right number for C, and the v1.0 slip was precisely that this contribution was made to do double duty and mislabelled as S.

4.4. **Numerical confirmation.** The identity (17) is confirmed numerically to 46 significant digits by the contour evaluation (Method B) and the Stokes-data evaluation (Method C) of Section 5; the differential-equation route (Method A) confirms the same identity *exactly* at the operator level (no numerical integration). The worst relative error across the numerical methods is 9.31×10^{-46} (`PERIOD-REP-VQUAD-003`, `numerical-integral.md` and `cross-verification.md`). We turn to the three verifications next.

5. Three verifications

We verify (17) three structurally independent ways: an exact differential-equation/operator computation (Method A, §5.2), a contour evaluation producing the Γ-factor in closed form (Method B, §5.3), and a Stokes-data computation that never touches γ (Method C, §5.4). The three use disjoint inputs and agree to 46 digits (§5.5). Sources: the slot `PERIOD-REP-VQUAD-003` deliverables `method-A/B/C-verification.md` and `cross-verification.md`, with scripts named inline.

5.1. **Strategy: why three methods, and why they are independent.** A single numerical coincidence to 46 digits is suggestive but not a proof; what upgrades it is that the three checks draw on *disjoint* mathematical inputs, so a hidden error in any one construction cannot be common to all three.

- **Method A** (differential/operator) uses only the two operators L_φ, L_V and the algebraic duality (23). It never evaluates an integral and never uses the numerical value of C; it certifies that the parameter-deformed integral $I_\gamma(z)$ *solves* L_φ, fixing the differential structure of the period.
- **Method B** (Borel–Laplace/contour) uses the explicit Hankel contour (19) and the local branch exponent $-(1+\beta)$ at $-\xi_0$. It produces the closed-form leading period $S\,e^{-\xi_0}$ analytically, using the analytic structure of $\widehat{B}$ but *not* the operators.

- **Method C** (Stokes/Galois) uses only the Stokes multiplier relation $S_{\text{mult}} = 2\pi i\, A/\Gamma(1+\beta)$ and the Galois-theoretic normalisation of the formal monodromy. It never touches γ at all, and is the tightest (9.31×10^{-46}).

Thus Method A certifies *which* differential equation the period solves, Method B certifies the *value* by a contour, and Method C certifies the *same value* by representation theory. Agreement of all three is the cross-check recorded in Table 2. (Sources: `cross-verification.md`.)

5.2. **Method A: differential-equation / operator duality.** Method A proves, *without any numerical integration*, that the parameter-deformed integral

$$I_\gamma(z) = \int_\gamma e^{-\xi/z}\, \widehat{B}(\xi)\, d\xi \tag{22}$$

is annihilated by L_φ, hence is a genuine solution of the V_quad linear equation whose connection coefficient is C.

The Borel sum of V_quad is $\varphi(z) = a_0 + \int_0^\infty e^{-\xi/z}\widehat{B}(\xi)\, d\xi$ with kernel $e^{-\xi/z}$, forced by the normalisation $b_m = a_{m+1}/m!$ (since $a_{m+1} = b_m\, m! = b_m \int_0^\infty e^{-t} t^m\, dt$). Along a contour with vanishing boundary terms—the rapid-decay thimble γ qualifies (Proposition 3.2)—the Laplace transform $\mathcal{L}[f](z) = \int e^{-\xi/z} f(\xi)\, d\xi$ intertwines the two differential structures by

$$D_\xi \;\longmapsto\; +\tfrac{1}{z}, \qquad \xi \;\longmapsto\; +z^2 D_z, \tag{23}$$

from $\partial_z e^{-\xi/z} = (\xi/z^2) e^{-\xi/z}$ and one integration by parts.

Proposition 5.1 (Operator duality)**.** *Write $L_V = \sum_{k,a} c_{k,a}\, \xi^a D_\xi^k$ with $c_{k,a} = [\xi^a] p_k$. The duality (23) sends L_V to the operator $M = \sum_{k,a} c_{k,a}\, (z^2 D_z)^a (1/z)^k$ acting on $I_\gamma(z)$. Since $\max_k \deg_\xi p_k = 2$, M has order 2, and over $\mathbb{Q}(\sqrt{3})$*

$$M \;=\; h(z)\, L_\varphi, \qquad h(z) = \frac{27\,(649 + 30\sqrt{3})}{418501\, z^2\, (2\sqrt{3} - 3)}, \tag{24}$$

the three coefficient ratios $M_{[D^2]}/q_2 = M_{[D^1]}/q_1 = M_{[D^0]}/q_0 = h(z)$ coinciding exactly. Hence $L_\varphi I_\gamma = 0$.

The computation is exact in $\mathbb{Q}(\sqrt{3})$ (script `stage4a_methodA_v2.py` → `stage4_methodA_results.json`). Its force comes from an anti-fluke test: the four sign conventions $(\pm 1/z, \pm z^2 D_z)$ were all tried, and *only* the correct Borel-sum convention ($D_\xi \mapsto +1/z$, $\xi \mapsto +z^2 D_z$) yields a proportional operator; the other three do not produce any $h(z)$. The full four-way enumeration is in Appendix A.4; it is the question every reader will ask, and the answer rules out an accidental match.

To see concretely why M has order 2, track the degrees. A monomial $c_{k,a}\, \xi^a D_\xi^k$ of L_V (with $a \le 2$, $k \le 4$) maps under (23) to $c_{k,a}\, (z^2 D_z)^a (1/z)^k$, an operator in z of order equal to $a \le 2$; summing over the four values of k at each fixed a collapses the $D_z^{>2}$ contributions, because the top-ξ-degree part of L_V (the $a = 2$ coefficients of p_2, p_3, p_4, which by (10) and Appendix A.1 share the common factor

$(70092+3240\sqrt{3})/418501)$ is precisely the symbol whose dualisation builds the order-2 leading term $h(z)q_2(z)D_z^2$. The exact match of all three ratios in (24) is then a nontrivial identity in $\mathbb{Q}(\sqrt{3})[z]$, not a degree count: it is the operator-level shadow of Borel–Laplace duality between L_φ and L_V.

Finally, I_γ is the difference of the two lateral Borel sums (the median-summation framework of [15]), so it is the subdominant solution of L_φ at the irregular point: as $z\to 0^+$, $I_\gamma(z)\sim(\text{const})\,e^{-\xi_0/z}z^{\bullet}$, and that constant is the *connection coefficient* C (in the normalisation of Theorem 4.1)—the Stokes multiplier of L_φ reweighted by the algebraic Γ-factor $|\Gamma(\beta)|/2\pi$, $C=(|\Gamma(\beta)|/2\pi)\,S$, as Methods B and C make explicit. Method A thus fixes the *differential structure*; Methods B and C fix the *value*.

5.3. **Method B: Borel–Laplace contour (Hankel).** Method B deforms the Borel-sum ray onto the thimble γ and evaluates the branch integral in closed form. With $\eta=\xi+\xi_0$ and the local form (12), the discontinuity integral (13) is the Hankel loop (19), giving the leading period

$$\int_\gamma e^{\xi}\widehat{B}(\xi)\,d\xi\Big|_{\text{lead}}=S\,e^{-\xi_0},\qquad |A|=K\,\Gamma(1+\beta),\tag{25}$$

The cancellation that produces (25) is worth displaying, since it is the heart of the Γ-factor mechanism. Substituting $A=(S/2\pi i)\,\Gamma(1+\beta)$ and using (19),

$$\begin{aligned}\int_\gamma e^{\xi}A(\xi+\xi_0)^{-(1+\beta)}\,d\xi &= A\,e^{-\xi_0}\oint_H e^{\eta}\eta^{-(1+\beta)}\,d\eta = A\,e^{-\xi_0}\frac{2\pi i}{\Gamma(1+\beta)}\\ &= \frac{S\,\Gamma(1+\beta)}{2\pi i}\,e^{-\xi_0}\frac{2\pi i}{\Gamma(1+\beta)}=S\,e^{-\xi_0},\end{aligned}\tag{26}$$

the $\Gamma(1+\beta)$ cancelling *exactly*: the branch integral has manufactured the Γ-factor that the connection coefficient carries. Hence $C=(|\Gamma(\beta)|/2\pi)\,S=|\Gamma(\beta)|\,K$ by (20). Numerically (script `stage4_methods.py`, Method B block, → `stage4_methods_results.json`, `mpmath`) the closed form matches the directly summed Borel data with relative error

$$\text{rel. err}=8.84\times10^{-46}.\tag{27}$$

Method B is independent of the Stokes datum: it *derives* $S\,e^{-\xi_0}$ from the cycle and the branch exponent, rather than assuming S.

5.4. **Method C: Stokes-data.** Method C uses only the deposited Stokes constant $S=2\pi K$ and the branch amplitude A extracted from L_V's large-order data, and never integrates over γ. The Stokes multiplier of L_φ at the irregular point is

$$S_{\text{mult}}=2\pi i\,\frac{A}{\Gamma(1+\beta)},\qquad |S_{\text{mult}}|=2\pi K=S,\qquad C=\frac{|A|}{|\beta|}=|\Gamma(\beta)|\,K.\tag{28}$$

Numerically (script `stage4_methods.py`, Method C block, → `stage4_methods_results.json`) the magnitude $|S_{\text{mult}}|$ matches $2\pi K$ with relative error 8.84×10^{-46} and C matches $|\Gamma(\beta)|K$ with relative error 9.31×10^{-46}—the tightest of the three, and the one requiring no contour integration. The factor i in $S_{\text{mult}}=2\pi iK$ is the Stokes

phase; the deposited real $S = 2\pi K$ is its magnitude, so there is no inconsistency with [1, 2].

The amplitude A that enters (28) is not a free parameter: it is extracted from the same coefficient stream (7) by the large-order law $b_m \sim \dfrac{A}{\Gamma(1+\beta)} \dfrac{\Gamma(m+1+\beta)}{\xi_0^{m+1+\beta}}$ (Proposition 2.4, Remark 2.5). Already the modest ratios b_{m+1}/b_m of the exact rationals in (7) approach $1/\xi_0 = \sqrt{3}/2$ (the reciprocal of the nearest Borel singularity, with the alternating sign locating it on the negative axis at $-\xi_0$), and a Richardson/Borel–Padé acceleration of the full stream fixes $|A| = K\,\Gamma(1+\beta)$ to 46 digits (script `stage4_methods.py`; the raw extraction is in `PERIOD-REP-VQUAD-002`, `borel_pade_census.py`). Thus Method C closes a loop: the concrete $\mathbb{Q}(\sqrt{3})$-rationals at small m determine, through their asymptotics, the very Stokes constant whose magnitude is the deposited $S = 2\pi K$.

The algebraic-times-Γ factor $2\pi i/\Gamma(1+\beta)$ in (28) is Galois-equivariant, which is what makes Method C a genuine *Galois* computation rather than a numerical coincidence. Its two pieces are the two torus generators of G_V: the period $2\pi i$ is the Betti–de Rham comparison period of the exponential connection E^{ξ} at the irregular point (the exponential-torus generator), and $1/\Gamma(1+\beta)$ is the branch normalisation at $-\xi_0$ (the $\mathbb{G}_m$ generator attached to the irrational exponent $-(1+\beta)$). Their product is the unipotent Stokes entry relating the de Rham class $[\omega]$ to the rapid-decay Betti class $[\gamma]$; this is exactly the Galois-equivariant pairing identified in `galois-LV-verification.md` §3, here made numerically explicit. We return to its motivic meaning in Section 6.

5.5. **Cross-verification.**

Proposition 5.2 (Consistency)**.** *Methods A, B, C all confirm the single identity* (17)*; none contradicts. They use disjoint inputs—exact operators over* $\mathbb{Q}(\sqrt{3})$ *(A), the rapid-decay contour and branch exponent (B), the Stokes constant and large-order amplitude (C)—yet land on the same branch datum* A *and the same* Γ*-factor* $\Gamma(1+\beta)$*. The numerical agreement is to* 46 *digits, with worst relative error* 9.31×10^{-46}*.*

Method	Mechanism	Independent of	Agreement
A (diff.-eq.)	$M = h(z)L_\varphi$ exact over $\mathbb{Q}(\sqrt{3})$	numerical integration	exact (symbolic)
B (Borel–Laplace)	Hankel $\Rightarrow$ leading period $Se^{-\xi_0}$	the Stokes datum	8.84×10^{-46}
C (Stokes-data)	$\lvert S_{\text{mult}}\rvert = 2\pi K$, $C = \lvert A\rvert/\lvert\beta\rvert$	γ-integration	8.84–9.31×10^{-46}

TABLE 2. The three verifications of (17). Source: `cross-verification.md`, `PERIOD-REP-VQUAD-003`.

The kernel signs in A ($e^{-\xi/z}$) and B ($e^{+\xi}$) are the two complementary faces of the one Borel–Laplace duality, not a contradiction: B fixes the value, A fixes the ODE. The probability of a spurious triple coincidence at 46 digits, with the unique correct operator-sign convention of four, is negligible; we record the result as **verified**.

6. Application to Fresán–Jossen and conditional transcendence

We now interpret the verified identity (17) inside the Fresán–Jossen theory of exponential motives and deduce the conditional transcendence of C. We are deliberately explicit about *both* layers of conditionality: the Fresán–Jossen period conjecture itself, and a motivic-comparison hypothesis we label G-MOTGALOIS. Sources: [7], and the slot deliverables `fj-application.md`, `galois-LV-verification.md`, `fresan-jossen-axioms.md`.

6.1. C as an exponential period.

Proposition 6.1 (C is a period of an exponential motive)**.** *The datum* (X, f, ω, γ) *with*

$$X = \mathbb{A}^1_\xi \smallsetminus \{0, -\xi_0\} \text{ over } \mathbb{Q}(\sqrt{3}), \qquad f = -\xi,$$
$$\omega = \widehat{B}(\xi)\, d\xi, \qquad \gamma \text{ as in Definition 3.1},$$

satisfies the Fresán–Jossen axioms: ω *is algebraic over the number field* $\mathbb{Q}(\sqrt{3})$ *(Theorem 2.3),* f *is regular with* $df = -d\xi$ *nowhere zero (no finite critical point), and* γ *is a rapid-decay cycle (Proposition 3.2, Proposition 3.3). Consequently* $\int_\gamma e^{\xi}\widehat{B}\, d\xi$ *is a period of the exponential motive* $H^1\big(\mathbb{A}^1 \smallsetminus \{0, -\xi_0\}, \nabla\big)$ *attached to* (X, f, ω)*, and by* (17) *so is* $C = (|\Gamma(\beta)|/2\pi) \int_\gamma e^{\xi}\widehat{B}\, d\xi$.

The auxiliary Fresán–Jossen condition most easily overlooked—the structure of the critical locus of f—is here the most favourable possible: $f = -\xi$ is linear, so X carries *no* interior critical point and the motive is a single $E^{-f} \otimes$ (rank-4 algebraic) with no extra vanishing-cycle contributions; the only critical value at infinity is the slope-1 irregularity (Proposition 2.4).

6.2. The period conjecture and the Galois input.

To an exponential motive M, Fresán–Jossen attach a motivic Galois group $G_{\mathrm{mot}}(M)$, making the period torsor a G_{mot}-torsor; their period conjecture ([7], Conjecture 1.3.2) is the exponential-motives analogue of Grothendieck's:

$$\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}\big(\text{periods of } M\big) = \dim G_{\mathrm{mot}}(M). \tag{29}$$

The Fresán–Jossen category of exponential motives is Tannakian over $\mathbb{Q}$; an exponential motive M generates a Tannakian subcategory $\langle M \rangle$ with affine group scheme $G_{\mathrm{mot}}(M) = \underline{\mathrm{Aut}}^{\otimes}(\omega_M)$, and the comparison between the Betti (rapid-decay) and de Rham realisations is a torsor under $G_{\mathrm{mot}}(M)$ whose period matrix has the periods of M as entries. Conjecture (29) is the statement that this period torsor is connected—equivalently, that the periods are as algebraically independent as the Tannakian formalism permits. For the motive M of Proposition 6.1 the relevant realisations are computed by the differential systems L_φ, L_V.

The differential-Galois data of Sections 2–5 compute the de Rham realisation of (a quotient of) $G_{\mathrm{mot}}(M)$:

- L_φ (order 2, regular side) has differential Galois group $\mathrm{SL}_2(\mathbb{C})$ (Theorem 1.2);

- L_V (order 4, Borel dual) has Galois group G_V containing a torus $\mathbb{G}_m$—from the irrational branch exponent $-1+\sqrt{3}/9$ at $-\xi_0$, whose monodromy eigenvalue $e^{2\pi i\sqrt{3}/9}$ has infinite order—together with the exponential/Stokes data at ∞ (formal torus $\times$ nontrivial unipotent Stokes, the SL_2-dual structure);
- C is the Galois-equivariant pairing $\langle[\gamma]_{\mathrm{rd}},[\omega]_{\mathrm{dR}}\rangle$, with the exponential period $2\pi i$ of E^{ξ} and the branch normalisation $1/\Gamma(1+\beta)$ as the two torus generators (Method C, (28)).

Because G_V carries both a non-abelian SL_2-type part and a transcendental-monodromy torus $\mathbb{G}_m$ with irrational character, the motivic Galois group is positive-dimensional and acts nontrivially on the class pairing defining C: there is no 1-dimensional sub-torsor forcing C to be algebraic, and no algebraic relation of C with the base period 1 is visible.

Concretely, the period pairing between the rapid-decay Betti class $[\gamma]$ and the de Rham class $[\omega]=[\widehat{B}\,d\xi]$, together with the unit and the exponential-torus generators, assembles into the period matrix

$$P(M)=\begin{pmatrix}1 & 0\\ \dfrac{1}{\Gamma(1+\beta)} & 2\pi i\end{pmatrix},\qquad \det P(M)=2\pi i, \tag{30}$$

whose entries are exactly the constants appearing in Method C, (28): the determinant $2\pi i$ is the exponential-torus period of E^{ξ}, the off-diagonal $1/\Gamma(1+\beta)$ is the branch normalisation at $-\xi_0$, and the connection coefficient $C=|\Gamma(\beta)|K=|A|/|\beta|$ is the period pairing assembled from these entries via the branch amplitude $|A|=K\,\Gamma(1+\beta)$, not itself an entry of $P(M)$. The transcendence of C is the statement that this matrix is not gauge-equivalent over $\overline{\mathbb{Q}}$ to a block-diagonal (algebraic) one—precisely what the period conjecture controls.

6.3. **The comparison gap G-MOTGALOIS.** The identification of the de Rham realisation G_V with the relevant quotient of $G_{\mathrm{mot}}(M)$ uses the standard de Rham-realisation comparison; the full Nori/Ayoub exponential-motive comparison for this *specific* M is *assumed*, not verified here.

> **Hypothesis (G-MOTGALOIS).** The de Rham/differential Galois group G_V computed in Sections 2–5 represents the motivic Galois group $G_{\mathrm{mot}}(M)$ of the exponential motive $M=(X,f,\omega)$ of Proposition 6.1; in particular $\dim G_{\mathrm{mot}}(M)\geq 1$ and C is not fixed by $G_{\mathrm{mot}}(M)$.

This is a conjectural bridge; it affects only the motivic *interpretation*, not any of the differential or numerical computations above. We flag it explicitly so that the transcendence corollary is never mistaken for unconditional.

6.4. **Conditional transcendence.**

Proof of Corollary 1.4. By Proposition 6.1, C is a period of M. Assume *(i)* the Fresán–Jossen period conjecture (29) and *(ii)* the hypothesis G-MOTGALOIS. By (ii), $\dim G_{\mathrm{mot}}(M)\geq 1$ and C is not fixed by $G_{\mathrm{mot}}(M)$, so C does not lie in the

algebraic (G_{mot}-invariant) part of the period algebra. By (i), every $\mathbb{Q}$-polynomial relation among the periods of M is of motivic origin, so a non-invariant period satisfies no algebraic relation over $\overline{\mathbb{Q}}$. Hence C is transcendental over $\overline{\mathbb{Q}}$. □

Remark 6.2 (The conjecture, specialised)**.** Concretely, Proposition 6.1 and the Galois input make the period algebra of M contain 1, $2\pi i$, $\Gamma(1+\beta)$ and C; the de Rham realisation G_V is positive-dimensional (it contains the torus $\mathbb{G}_m$ acting through the irrational character $\sqrt{3}/9$ and an SL_2-type Stokes part). The period conjecture (29) predicts $\mathrm{trdeg}_{\mathbb{Q}}$ of this algebra equals $\dim G_{\mathrm{mot}}(M)$, which under G-MOTGALOIS is ≥ 1 with C outside the invariants; equality at any value ≥ 1 already forces $C \notin \overline{\mathbb{Q}}$. The statement is thus robust to the precise value of $\dim G_{\mathrm{mot}}(M)$: only positivity and non-invariance of C are used, not the exact dimension.

Remark 6.3 (What is unconditional, and what is not)**.** Independently of Fresán–Jossen, $C = |\Gamma(\beta)|\,K$ with $\beta = -1/(3\sqrt{3}) \notin \mathbb{Q}$. The factor $|\Gamma(\beta)|$ is a value of Γ at an algebraic-irrational argument (transcendence of such values is itself only known conditionally in general; [25] settles the classical rational-argument and modular cases), while the transcendence of K remains conjectural. Thus the *product* C is only *expected* transcendental by ad-hoc Γ-arithmetic. The value of the Fresán–Jossen route is that it upgrades the *whole* of C to a single structured conditional statement, tied to the motivic Galois group rather than to Γ-arithmetic of one factor. We do not collapse this into an unconditional claim.

7. Discussion

7.1. **Place in the Sakai stratification.** The master conjecture of the Sakai-stratification program [3] predicts, for each Sakai surface stratum, that the connection constant of a polynomial continued fraction whose tail is governed by the corresponding Painlevé equation is an exponential period, and—in part (ii)(a)—that for the $D_5^{(1)}$ (Painlevé V) stratum this period is explicit. For V_{quad}, the rank-one $D_5^{(1)}$ representative, that entry was previously graded *structural*: the skeleton $C = \Gamma(\beta)\cdot K$ matched, but no period integral was exhibited. Theorem 1.1 exhibits the integral and verifies it three ways, so the present paper *completes part (ii)(a) for* $d = 2$: the $D_5^{(1)}$ connection constant is realised as an explicit Fresán–Jossen exponential period, and (conditionally) transcendental. We do not modify the live stratification deposit; the upgrade is recorded as a recommendation (`PERIOD-REP-VQUAD-003`, `ledger.json`). Adjacent instances in the same program corroborate the pattern: the Edge–Borel ring papers characterise the connection/amplitude datum of related polynomial-continued-fraction families [4, 5], and a companion Fredholm-determinant representation realises a related pcf growth constant as an explicit determinant [6].

7.2. **The $d \geq 3$ program.** The method is structural rather than special to $d = 2$, but two ingredients become harder for the higher strata $d \geq 3$ (Painlevé VI/Garnier and degenerations). First, the exact coefficient field: for V_{quad} the Riccati seed $\sigma = -1/\sqrt{3}$ lands in $\mathbb{Q}(\sqrt{3})$, which is what makes L_φ, L_V algebraic over a *real quadratic* field; for $d \geq 3$ the analogous seed need not be quadratic, and holonomic recognition may

require a higher-degree number field (or fail to be holonomic at all). Second, the connection problem: the single-branch structure at $-\xi_0$ that makes the thimble see the entire connection datum is a feature of the finite resurgence (Corollary 1.3); higher strata may carry several branches and a genuinely multi-dimensional Stokes structure. The methodology suggests the right first test at each d is the *holonomicity-and-field* question for the Borel transform, exactly as in `PERIOD-REP-VQUAD-002`; only if that passes does the period extraction proceed.

7.3. **Relation to the Ramanujan Machine and conservative matrix fields.** V_{quad} is a polynomial continued fraction of the kind systematically generated by the Ramanujan Machine [20] and organised by the conservative-matrix-field formalism [21]. Those frameworks excel at *discovering* continued-fraction identities and at conjecturing the *value* of a convergent; the object here is complementary and orthogonal: the connection coefficient C is a *divergent*-series (Stokes/Borel) datum, invisible to convergence-rate heuristics, and the content of this paper is its *period* structure rather than a closed-form value. We see the two as a natural pairing—machine discovery of the continued fraction, motivic analysis of its transcendence fingerprint.

7.4. **Relation to topological recursion.** The Stokes data and isomonodromic τ-functions of Painlevé equations are computed, for broad classes of spectral curves, by the Eynard–Orantin topological recursion and the isomonodromic Hamiltonian/reconstruction program [22, 23, 24]. That program produces the τ-series and the Stokes constants of a Painlevé transcendent from the geometry of a spectral curve; the holonomic/Borel route taken here is complementary. It does not reconstruct the τ-function: it isolates the minimal scalar operator L_V with an exact $\mathbb{Q}(\sqrt{3})$ certificate and extracts the *period* of the Stokes datum—the exponential period (17)—which is the object of arithmetic interest here. The two viewpoints meet at the Stokes constant $S = 2\pi K$: topological recursion would compute it from a spectral curve, whereas we read it from the rapid-decay period (18) and reweight it to the connection coefficient C by the algebraic Γ-factor $|\Gamma(\beta)|/2\pi$.

7.5. **Comparison with established period constants.** It is instructive to place C among known periods. Classical (convergent) periods—π, $\log 2$, $\zeta(3)$, values of hypergeometric integrals—are periods of *ordinary* motives, captured by Kontsevich–Zagier integrals of algebraic forms over algebraic domains. The constant C is not of this kind: it is a *Stokes* constant, attached to the divergence of a Gevrey-1 series, and its integral representation requires the *exponential* factor e^{ξ} and a rapid-decay (non-compact) cycle—precisely the data Fresán–Jossen add to the classical picture. In this sense C sits alongside the exponential periods $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$ and the Bessel/Airy connection constants, rather than alongside $\zeta(3)$. What is unusual here is the provenance: C is manufactured by a *number-theoretic* object (a polynomial continued fraction), yet its period structure is governed by a real-quadratic algebraic connection. The single Γ-value $\Gamma(\beta)$ visible in (17) is the trace of the exponential nature of the period, exactly as $\Gamma(s)$ is itself the prototypical exponential period.

7.6. **What computer algebra settles, and what it does not.** It is worth stating precisely where automation ends, both to orient the reader and because a computer-algebra audience will reasonably ask. Present symbolic software *does* settle the differential backbone: holonomic recognition of L_φ and L_V from the coefficient stream is `gfun`/`ore_algebra` guessing [18, 19] followed by an exact residual certificate over $\mathbb{Q}(\sqrt{3})$; the differential Galois group is decided by Kovacic's algorithm [8] (and by packages such as Maple's `DEtools[DifferentialGaloisGroup]`); the large-order/Borel–Padé extraction of K, S and the branch exponent is standard exponential asymptotics. None of these steps is conjectural, and we have used all of them. What computer algebra does *not* settle is exactly the mathematical content: (a) the identification of the precise rapid-decay cycle γ and the *proof* that the three period-extraction routes coincide (Section 5); and (b) the motivic interpretation and the conditional transcendence (Corollary 1.4), which lie beyond any current CAS and rest on the Fresán–Jossen conjecture and the hypothesis G-MOTGALOIS. A CAS can confirm every algebraic identity in this paper; it cannot supply the period-theoretic meaning. The reproducibility statement (Appendix A.5) lists the exact tools and versions.

7.7. **Open problems.**

(1) **G-MOTGALOIS.** Verify the Nori/Ayoub exponential-motive comparison for $M = (\mathbb{A}^1 \smallsetminus \{0, -\xi_0\}, -\xi, \widehat{B}\, d\xi)$, removing the second conditionality of Corollary 1.4.

(2) **The $d \geq 3$ connection problem.** Determine the coefficient field and the branch structure of the Borel transform for the Painlevé-VI/Garnier strata, and decide whether the single-branch period extraction survives.

(3) **Functoriality.** Is the assignment (PCF stratum) $\mapsto$ (exponential motive M) functorial along the Sakai degeneration cascade, so that confluences of Painlevé equations induce maps of exponential motives compatible with the connection constants?

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Conflict of interest

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Data availability

The data, verification code, and exact-computation certificates supporting the findings of this article are openly available in the VQUAD-PERIODREP reproducibility bundle deposited in Zenodo, concept DOI `10.5281/zenodo.20719042`, which resolves to the latest version; see Appendix A.5 for details.

Declaration on the use of AI tools

The research and the preparation of this manuscript made use of AI assistants (large language models) under the author's SIARC/AEAL governance framework. All mathematical claims were independently verified by the exact-computation certificates described in Appendix A.5. The author takes full responsibility for the content of this article.

Appendix A. Explicit operators, Kovacic certificate, logs, conventions, reproducibility

A.1. **Full coefficient listings.** The operator $L_\varphi = q_2D^2 + q_1D + q_0$ is given in full in Theorem 2.1. Here we record the Borel-dual operator $L_V = \sum_{k=0}^{4} p_k(\xi)D^k$, normalised by $p_0 \equiv 1$, with coefficients in $\mathbb{Q}(\sqrt{3})$ (denominators 431 and $418501 = 431 \times 971$):

$$
\begin{aligned}
p_0 &= 1,\\
p_1 &= \frac{659+150\sqrt{3}}{431} + \frac{432+12\sqrt{3}}{431}\,\xi,\\
p_2 &= \frac{2552175+199224\sqrt{3}}{418501} + \frac{496044+61620\sqrt{3}}{418501}\,\xi + \frac{70092+3240\sqrt{3}}{418501}\,\xi^2,\\
p_3 &= \frac{77760+560736\sqrt{3}}{418501} + \frac{1685448+101124\sqrt{3}}{418501}\,\xi + \frac{70092+3240\sqrt{3}}{418501}\,\xi^2,\\
p_4 &= \frac{19440+140184\sqrt{3}}{418501}\,\xi + \frac{210276+9720\sqrt{3}}{418501}\,\xi^2 = \frac{210276+9720\sqrt{3}}{418501}\,\xi\Big(\xi+\tfrac{2}{\sqrt{3}}\Big).
\end{aligned}
$$

The residual $L_V\widehat{B}$ vanishes identically in $\mathbb{Q}(\sqrt{3})$ through order ξ^{129} (`stage0_residual_check.py`, method A: `sympy` exact over $\mathbb{Q}(\sqrt{3})$; method B: `mpmath` at 160 digits, cross-checked). Source: `PERIOD-REP-VQUAD-002`, `operator-verification.md` §4.0.

A.2. **Kovacic certificate for L_φ.** Reducing L_φ to $u'' = r\,u$ via $y = u\exp(-\frac{1}{2}\int q_1/q_2)$ gives (8),

$$r(z) = \frac{11z^4+4z^2+4z+12}{4z^4(z^2+z+3)^2},$$

with pole orders $\{0:4,\ \rho_\pm:2\}$, $\rho_\pm = \frac{-1\pm i\sqrt{11}}{2}$, $o(\infty) = 4$, and leading Laurent coefficient $\frac{1}{3}z^{-4}$ at 0. Kovacic case elimination (`stage2_kovacic.py`, `stage2b_symsquare.py`):

- *Case 3* (finite groups): requires every pole order ≤ 2; the order-4 pole at $z = 0$ excludes it.
- *Case 1* (reducible): requires a rational solution of $v' = r - v^2$; the rational Riccati solver returns the empty set.
- *Case 2* (imprimitive): with Case 1 excluded, requires a rational solution of the symmetric square $L_\varphi^{\odot 2} = D^3 - 4rD - 2r'$; the ansatz $f = N(z)/(z^8(z^2+z+3)^4)$, $\deg N \le 18$, yields a homogeneous 38-unknown $\mathbb{Q}$-linear system with only the trivial solution.
- *Case 4*: by elimination, $G = \mathrm{SL}_2(\mathbb{C})$.

Independent structural confirmation: the reduced equation has no first-order term, so $G \subseteq \mathrm{SL}_2$; the two distinct exponentials $\exp(\pm(1/\sqrt{3})/z)$ at $z = 0$ generate the diagonal torus, and the nonzero Stokes constant $S = 2\pi K$ supplies an off-torus unipotent; torus-plus-unipotent generate SL_2. Source: `PERIOD-REP-VQUAD-003`, `kovacic-verification.md`.

A.3. **Numerical logs.** All constants to the stated precision (`PERIOD-REP-VQUAD-001`, `numerical-check.md`; `mpmath`):

$$K = 0.072878102551866964129442363329652512804555689 2\ldots \quad (58 \text{ digits}),$$
$$S = 2\pi K = 0.457906623169017636119097842548225837962395135\ldots,$$
$$\beta = -1/(3\sqrt{3}) = -0.19245008972987525\ldots, \qquad \xi_0 = 2/\sqrt{3} = 1.1547005383792517\ldots,$$
$$C = |\Gamma(\beta)|K = 0.437705286193537221230739749794369589981725597\ldots.$$

Method agreement (`stage4_methods.py`, `stage1_hankel_period.py` at dps 160–260): leading thimble period $= S\,e^{-\xi_0}$ to relative error 8.84×10^{-46} (Method B); $|S_{\mathrm{mult}}| = 2\pi K$ to 8.84×10^{-46} and $C = |\Gamma(\beta)|K$ to 9.31×10^{-46} (Method C). Frobenius solution at $-\xi_0$: exponents $\{-(1+\beta), 0, 1, 2\}$, no logarithmic terms, recurrence residual 1.6×10^{-46} (`stage3b_frobenius_v2.py`).

A.4. **The four sign conventions for Method A.** Method A (§5.2) hinges on the Borel–Laplace intertwiner (23). Because both the kernel exponent and the ξ-multiplication carry a sign, there are four a priori conventions; we tested all four (`stage4a_methodA_v2.py`) and only one produces a proportional operator $M = h(z)L_\varphi$:

$D_\xi \mapsto$	$\xi \mapsto$	kernel	result
$+1/z$	$+z^2 D_z$	$e^{-\xi/z}$ (Borel sum)	$M = h(z)L_\varphi$, $h = \dfrac{27(649+30\sqrt{3})}{418501\, z^2(2\sqrt{3}-3)}$
$+1/z$	$-z^2 D_z$	—	three coefficient ratios disagree
$-1/z$	$+z^2 D_z$	—	three coefficient ratios disagree
$-1/z$	$-z^2 D_z$	$e^{+\xi/z}$	three coefficient ratios disagree

Only the kernel forced by the normalisation $b_m = a_{m+1}/m!$ (top row) works. This is the anti-fluke test: a spurious operator proportionality would not have singled out the correct analytic convention. We record it here because every reader reconstructing Method A will face the same sign choice. Source: `PERIOD-REP-VQUAD-003`, `method-A-verification.md`.

A.5. **Reproducibility statement and code availability.** Every computational claim of this paper is reproducible from the scripts in the probe slots `PERIOD-REP-VQUAD-001/002/003` (SIARC relay bridge). The differential-algebra is exact over $\mathbb{Q}(\sqrt{3})$ (`sympy` 1.14.0); the asymptotic/period numerics use `mpmath` 1.3.0 at 160–260 decimal digits; Python 3.12.10. The parent V_quad deposit is Zenodo concept DOI `10.5281/zenodo.20455089` [1], with the $S = 2\pi K$ calibration in version 1.2, `10.5281/zenodo.20481592` [2]. Key scripts: `stage0_residual_check.py` (operator residuals), `stage2_kovacic.py` and `stage2b_symsquare.py` (Galois), `stage3b_frobenius_v2.py` (branch exponent), `stage4a_methodA_v2.py` (Method A and the four-convention test), `stage4_methods.py` (Methods B, C), `stage1_hankel_period.py` (Hankel period). The LaTeX source compiles with `pdflatex` (MiKTeX 25.12) under a fixed `SOURCE_DATE_EPOCH` for byte-reproducible output. No proprietary software is required; the Kovacic verdict, optionally cross-checkable in Maple's `DEtools[DifferentialGaloisGroup]`, is here obtained by open case-elimination plus the structural argument of Appendix A.2.

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